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GATE ECE · ANALOG CIRCUITS · CHAPTER-WISE PYQS

# Feedback Amplifier

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

| 19 | 17 | 1989 – 2024 | English |
|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | LANGUAGE |

INSIDE THIS PDF

| | |
|---|---|
| Marks 1 | 10 questions |
| Marks 2 | 9 questions |
| Answer key | at the end |

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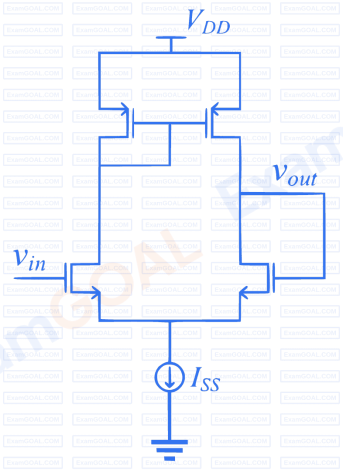
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3. **Understand**
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## Marks 1 10 questions

**Q1.** GATE ECE 2024

Open solution

For the closed loop amplifier circuit shown below, the magnitude of open loop low frequency small signal voltage gain is 40. All the transistors are biased in saturation. The current source $I_{ss}$ is ideal. Neglect body effect, channel length modulation and intrinsic device capacitances. The closed loop low frequency small signal voltage gain $\frac{v_{out}}{v_{in}}$ (rounded off to three decimal places) is ____.




(A) 0.976 (B) 1.000 (C) 1.025 (D) 0.488

**Q2.** GATE ECE 2023

Open solution

A cascade of common-source amplifiers in a unity gain feedback configuration oscillates when

(A) the closed loop gain is less than 1 and the phase shift is less than 180°.

(B) the closed loop gain is greater than 1 and the phase shift is less than 180°.

(C) the closed loop gain is less than 1 and the phase shift is greater than 180°.

(D) the closed loop gain is greater than 1 and the phase shift is greater than 180°.

**Q3.** GATE ECE 2014 Set 1

Open solution

A good current buffer has

(A) low input impedance and low output impedance.

(B) low input impedance and high output impedance.

(C) high input impedance and low output impedance.

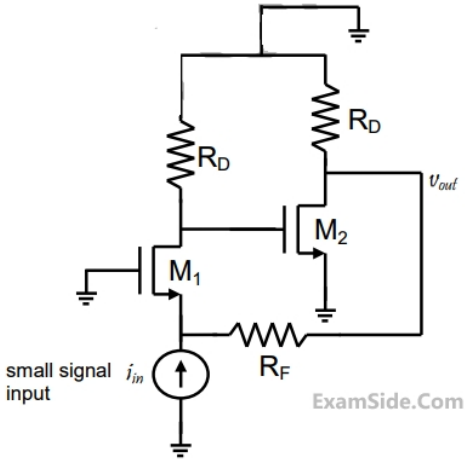
(D) high input impedance and high output impedance.

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**Q4.** GATE ECE 2014 Set 1

Open solution

In the ac equivalent circuit shown in the figure, if $i_F$ is very large, the type of feedback is

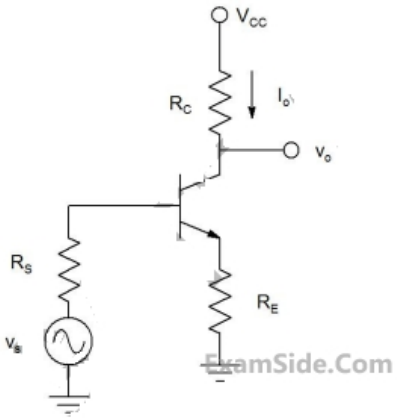



(A) voltage-voltage feedback
(B) voltage-current feedback
(C) current-voltage feedback
(D) current-current feedback

**Q5.** GATE ECE 2014 Set 2

Open solution

The feedback topology in the amplifier circuit (the base bias circuit is not shown for simplicity) in the figure is




(A) Voltage shunt feedback.
(B) Current series feedback.
(C) Current shunt feedback.
(D) Voltage series feedback.

**Q6.** GATE ECE 2014 Set 3

Open solution

The desirable characteristics of a transconductance amplifier are

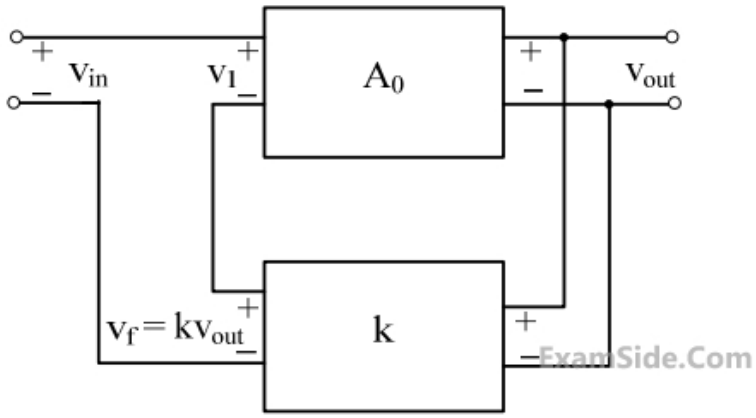
(A) high input resistance and high output resistance
(B) high input resistance and low output resistance
(C) low input resistance and high output resistance
(D) low input resistance and low output resistance

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**Q7.** GATE ECE 2013

In a voltage-voltage feedback as shown below, which one of the following statements is TRUE, if the gain k is increased?

Open solution




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(A) The input impedance increases and output impedance decreases.

(B) The input impedance increases and output impedance also increases.

(C) The input impedance increases and output impedance also increases.

(D) The input impedance increases and output impedance increases.

**Q8.** GATE ECE 1998

In a shunt-shunt negative feedback Amplifier, as compared to the basic Amplifier,

Open solution

(A) both input and output impedance decreases .

(B) input impedance decreases but output impedance increases.

(C) input impedance increases but output impedance decreases

(D) both input and output impedance decreases

**Q9.** GATE ECE 1995

To obtain very high input and output impedances in a feedback Amplifier, the mostly used is

Open solution

(A) Voltage - series (B) Current - series (C) Voltage - shunt (D) Current - shunt

**Q10.** GATE ECE 1993

Negative feedback in Amplifiers.

Open solution

(A) improves the signal to noise ratio at the i/p.

(B) improves the signal to noise ratio at the o/p.

(C) does not effect the signal to noise ratio at the o/p.

(D) Reduces distortion

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## Marks 2 9 questions

**Q11.** GATE ECE 2004

Voltage series feedback (also called series-shunt feedback) results in

(A) increase in both input and output impedances.

(B) decrease in both input and output impedances.

(C) increase in both input impedances and decrease in output impedances.

(D) decrease in input impedances and increase in output impedance.

Open solution

**Q12.** GATE ECE 2003

An amplifier without feedback has a voltage gain of 50, input Resistance of 1 kΩ and output Resistance of 2.5 kΩ . The input Resistance of the current-shunt negative feedback factor of 0.2 is

(A) 1/1 kΩ (B) 1/5 kΩ (C) 5 kΩ (D) 11 kΩ

Open solution

**Q13.** GATE ECE 2002

In a negative feedback amplifier using voltage - Series (i.e., voltage - sampling, series mixing) feedback

(A) $R_i$ decreases and $R_0$ decreases
($R_i$ and $R_0$ denote the input and outout resistances respectively)

(B) $R_i$ decreases and $R_0$ increases
($R_i$ and $R_0$ denote the input and outout resistances respectively)

(C) $R_i$ increases and $R_0$ decreases
($R_i$ and $R_0$ denote the input and outout resistances respectively)

(D) $R_i$ increases and $R_0$ increases
($R_i$ and $R_0$ denote the input and outout resistances respectively)

Open solution

**Q14.** GATE ECE 2000

The most commonly used Amplifier in sample and hold circuit is

(A) A unity gain inverting Amplifier

(B) A unity gain non-inverting inverting Amplifier

(C) An inverting Amplifier with a gain of 10.

(D) An inverting Amplifier with a gain of 100.

Open solution

**Q15.** GATE ECE 1999

Negative feedback in an amplifier

(A) Reduces gain (B) Increase frequency and phase distortions

(C) Reduces bandth (D) Increase Noise

Open solution

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**Q16.** GATE ECE 1999

An amplifier has an open loop gain of 100, an input impedance of 1 $k_{\Omega}$, and an output impedance of 100 $_{\Omega}$ . A feedback network with a feedback factor of 0.99 is connected to the amplifier an a voltage series feedback mode. The new input and output impedance respectively are

Open solution

(A) $10\,\Omega\,and\,1\,\Omega$ (B) $10\,\Omega\,and\,10\,\Omega$ (C) $100k\,\Omega\,and\,1\,\Omega$ (D) $100k\,\Omega\,and\,1\,\Omega$

**Q17.** GATE ECE 1997

Negative feedback in
1. Voltage series configuration
2.Current shunt configuration

(a) increases input impedance
(b) decreases input impedance
(c) increases closed loop gain
(d) leads to ascillation.

Open solution

**Q18.** GATE ECE 1991

Two non-inverting amplifiers, one having a unity gain and other having a gain of twenty, are made using identical operational gain amplifier, the amplifier with gain twenty has

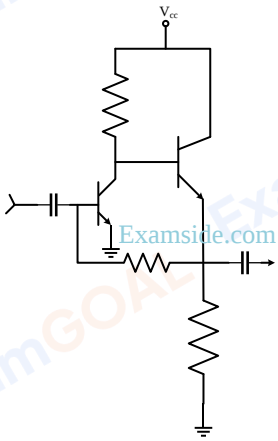
Open solution

(A) less negative feedback (B) greater input impedance
(C) less bandwidth (D) none of the above

**Q19.** GATE ECE 1989

The feedback amplifier shown in Fig.has:

Open solution




(A) current - series feedback with large input impedance and large output impedance.
(B) voltage - series feedback with large input impedance and low output impedance.
(C) voltage - shunt feedback with low input impedance and low output impedance.
(D) current - shunt feedback with low input impedance and output impedance.

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# Answer Key

## Marks 1

| | | | | |
|---|---|---|---|---|
| 1. **A** | 2. **D** | 3. **A** | 4. **B** | 5. **B** |
| 6. **A** | 7. **A** | 8. **A** | 9. **B** | 10. **B, D** |

## Marks 2

| | | | | |
|---|---|---|---|---|
| 11. **C** | 12. **B** | 13. **C** | 14. **B** | 15. **A** |
| 16. **C** | 17. **1-(a) , 2-(b)** | 18. **A, C** | 19. **D** | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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